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Evaluation of the TEOM® Method for Measurement of Ambient Particulate Mass in Urban Areas

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ABSTRACT

Increased interest in the health effects of ambient particulate mass (PM) has focused attention on the evaluation of existing mass measurement methodologies and the definition of PM in ambient air. The Rupprecht and Patashnick Tapered Element Oscillating MicroBalance (TEOM®) method for PM is compared with time-integrated gravimetric (manual) PM methods in large urban areas during different seasons. Comparisons are conducted for both PM_{10} and $PM_{2.5}$ concentrations.

In urban areas, a substantial fraction of ambient PM can be semi-volatile material. A larger fraction of this component of PM_{10} may be lost from the TEOM-heated filter than the Federal Reference Method (FRM). The observed relationship between TEOM and FRM methods varied widely among sites and seasons. In East Coast urban areas during the summer, the methods were highly correlated with good agreement. In the winter, correlation was somewhat lower, with TEOM PM concentrations generally lower than the FRM. Rubidoux, CA, and two Mexican sites (Tlalnepantla and Merced) had the highest levels of PM_{10} and the largest difference between TEOM and manual methods.

$PM_{2.5}$ data from collocation of 24-hour manual samples with the TEOM are also presented. As most of the semi-volatile PM is in the fine fraction, differences between these methods are larger for $PM_{2.5}$ than for PM_{10}.

IMPLICATIONS

Semi-volatile organic and inorganic compounds can represent a substantial fraction of ambient particulate mass. Loss of semi-volatile compounds during filter collection can occur due to changes in the composition of the air sample. Heating of the air sample to a temperature 10-20 °C above ambient to remove particle-bound water can further enhance loss of the semi-volatile constituents of ambient particles. These losses depend on particle composition, which varies with location and time. Thus, it is not possible to determine the true concentration by using empirical factors to correct the observed concentrations. Therefore, to minimize losses of semi-volatile compounds, we recommend that particle mass measurements be made at temperatures comparable to ambient.

INTRODUCTION

Several studies over the last few years have shown significant relationships between PM_{10} (concentration of particulate mass in ambient air with an aerodynamic diameter of less than 10 µm) and human health effects, including mortality and morbidity, for levels at and below current national ambient air quality standards for PM_{10}.[1-3] Some of these studies have implied that the fine mass ($PM_{2.5}$) component of PM_{10}, which is primarily from combustion products of fossil fuels, is more likely to be associated with the observed health effects than the coarse fraction (2.5-10 µm) particles.[4] The results from these studies, and the U.S. Environmental Protection Agency's (EPA) accelerated timetable for conducting review of the particulate mass (PM) standards, have focused attention on potential differences between existing EPA-approved PM_{10} measurement methods, as well as the performance of possible candidates for a Federal Reference Method (FRM) for fine mass.

The presence of inorganic and organic semi-volatile material (SVM) in the particulate phase introduces a degree of difficulty and uncertainty in measuring atmospheric PM. Semi-volatile or volatile components, such as water associated with hygroscopic material (particle-bound water), ammonium nitrate in equilibrium with nitric acid and ammonia, and semi-volatile organic compounds (SVOCs), may account for a substantial fraction of the total particulate mass observed in certain locations. The amount of semi-volatile material associated with the particulate phase is a function of temperature, relative humidity, and its gas-phase concentration.

In many instances, most of the adsorbed water may be removed during the equilibration protocol for determining the mass of fine particles. The FRM calls for collection of PM on a filter and, prior to weighing, equilibration of the filter and collected PM at a temperature of 15-30 °C, with a relative humidity of less than 50%. This treatment removes most of the particle-bound water (PBW). However, this process could lead to some loss of SVM after particle collection, including SVOC and ammonium nitrate, and these losses could vary significantly within the equilibration limits set by the FRM. More importantly, the ambient conditions of temperature and humidity that the sample is exposed to both during and after collection (before removal from the sampler) are completely uncontrolled, and can lead to additional variability in the amount of semi-volatile PM that is measured with the FRM.

The Rupprecht and Patashnick Tapered Element Oscillating MicroBalance (TEOM®) method provides continuous measurement of PM, and was designated by the EPA in October 1990 as an equivalent method for measurement of 24-hour mean PM_{10} for compliance purposes.[5] The TEOM is a frequently used and cost-effective method for measurement of PM_{10} when daily or hourly concentrations are needed for regulatory purposes, atmospheric chemistry studies, source apportionment modeling, or epidemiologic studies of human health effects. Along with an ability to provide near-continuous PM measurements, this method does not use any radioactive sources, and provides direct measurement of particulate mass collected on the filter.

In its standard configuration, the TEOM collects particles at 3 liters per minute (lpm) on an oscillating filter heated to 50 °C. Under certain conditions, lower temperatures and flow rates can be used to minimize loss of SVM;[12] however, all of the TEOM data presented here was collected at the flow and temperature noted above. The TEOM filter is heated to eliminate interference from PBW and to provide a stable and reproducible measurement of ambient PM_{10} without the SVM component. Removal of PBW is critical to the operation of the TEOM, which is based on the measure of incremental mass, because particulate matter is accumulated on the oscillating filter up to two or three weeks before a new filter is used. If the collected PM is not heated or desiccated sufficiently to remove all of the water, any change in the ambient water vapor concentration will produce a change in PBW, which will in turn change the collected PM on the TEOM filter.

Since much of the semi-volatile particulate matter is expected to be lost from the TEOM filter during and after collection at 50 °C, there is the potential for a substantially different measurement of PM_{10} with the TEOM than with the FRM. This is most likely to occur in urban areas (or areas affected by urban plumes) where nitrates and organics can comprise a substantial fraction of the PM_{10} mass. As discussed above, the FRM and other manual PM_{10} methods are also likely to have some loss of SVM. Because they collect the sample at lower temperatures than the TEOM, the manual methods will usually retain more semi-volatile PM than the TEOM. The differences between the methods would be expected to be primarily a function of the aerosol composition, which is a function of site and season of the year.

The purpose of this study was to evaluate the performance of the TEOM method by comparing it with collocated time-integrated samplers. Comparisons were conducted during different seasons in various U.S. and Mexican locations, where particulate composition was expected to differ. The performance evaluation was conducted by comparing both PM_{10} and $PM_{2.5}$ concentrations determined by TEOM to concentrations obtained with time-integrated samplers.

METHODS

The TEOM measures the accumulation of mass on a heated filter attached to the tip of a hollow, tapered, oscillating glass rod. The change in the oscillation frequency is used to make a direct measurement of the accumulation of mass on the filter over time. The flow rate through the filter is measured with a mass flow meter. PM concentration is determined every 2 seconds from these two measurements. The method is fully described elsewhere.[6]

For PM_{10} measurements, a Model 1400 or 1400A TEOM was used. Data were taken from the analog output, which was run with a range of -100 to +200 μg m^{-3} to avoid truncating negative instrument responses. The standard filter sample flow rate (31 lpm) and temperature (50 °C) were used. At this flow rate, the face velocity at the TEOM filter is 38 cm/s. In this configuration, the limit of detection (LOD) of the method is less than 2 μg m^{-3} for a 24-hour integrated sample, according to the manufacturer. All TEOM data are reported at standard temperature and pressure.

For $PM_{2.5}$, a Harvard-Marple Impactor (HI) inlet[7,8] was placed upstream from the TEOM to remove particles larger than 2.5 μm. Inlet flow rates were 4, 10, or 20 lpm, depending on the site. For the 4 lpm HI, no flow splitter was used, so the flow through the TEOM filter was 4 lpm. A flow splitter was used for the 10 and 20 lpm flow rates, resulting in a filter flow of 1.8 and 3.6 lpm, respectively. The face velocities at the TEOM filter were 22.7, 45.2, and 50 cm/s for 1.8, 3.6, and 4 lpm, respectively.

The TEOM PM_{10} concentrations (in μg m^{-3}) include the standard calibration factors (b_0 = 3 μg m^{-3} and b_1 = 1.03) as required by the FRM designation. The intercept (b_0) and slope (b_1) of a linear regression between TEOM and a collocated FRM PM_{10} measurement are used to compensate for the loss of PBW and SVM at the elevated temperature of TEOM sample collection.[6] Any adjustment of TEOM data for $PM_{2.5}$ would be different. Since it was not the goal of this work to derive appropriate $PM_{2.5}$ calibration factors, which would likely be dependent on the site and/or season, calibration factors of $b_0 = 0$ µg m^{-3} and $b_1 = 1.0$ were used for all the fine mass TEOM data reported.

Time-Integrated PM_{10} and $PM_{2.5}$

Two different methods were used for time-integrated PM_{10} measurements. The Sierra-Anderson Model 1200 Size Selective Inlet (SSI) Hi-Volume sampler with quartz fiber filters (an FRM for PM_{10}) was used at the California and Mexico City sites. Nitrate was also measured on these filters for the California sites. The Harvard-Marple Impactor sampler (HI) operating at 10 lpm was used at the Washington, D.C.; Philadelphia, PA; and Boston, MA, sites. Filters were weighed after equilibration at 40% RH. $PM_{2.5}$ was measured with the 10 lpm HI method except at Uniontown, PA, where the Harvard/EPA Annular Denuder System (HEADS) operating with a $PM_{2.1}$ inlet at 10 lpm was used.[9] All SSI and HI concentrations are corrected to standard temperature and pressure. The LOD for the HI and HEADS methods is 3 µg m^{-3}; for the SSI method, the LOD is 2 µg m^{-3}. The face velocities at the SSI, HI, and HEADS filters were 41, 20, and 12.1 cm/s, respectively.

The pressure drop across the filter can affect the degree of evaporation of volatile particulate matter collected on the filter. This pressure drop is a function of the filter face velocity. Since the filter face velocities for manual samplers are in the same range with those of the TEOM filter (Table 3), any difference in the particle collection characteristics of TEOM and manual samplers should be attributed to factors other than differences in the pressure drop across the sampler's filter.

Comparisons between the Harvard Impactor and EPA-approved SSI HiVol and Sierra Dichotomous PM_{10} and $PM_{2.5}$ sampling methods were made in Azusa, CA, in March 1989 for 12-hour samples. As shown in Table 1, the agreement between these methods is excellent.

Table 1. SSI-HiVol and dichotomous sampler vs. Harvard Impactor, Azusa, CA, March 1989, 12-hour samples.

PM_{10} Method	r^2	N	Mean HI/ Reference	Ratio of Means
SSI-HiVol	0.93	20	57/58	0.98
Sierra Dichot	0.97	18	59/61	0.97
$PM_{2.5}$ Method				
Sierra Dichot	0.96	19	41/42	0.98

(all r^2 significant at p = .05)

[1] Mean concentrations are in µg m^{-3}

Site Locations

Data for the method comparisons presented were collected at several urban and two non-urban sites on the East and West Coasts of the United States and Mexico, during different seasons of the year. Table 2 lists the sites, seasons, and particulate matter size fractions measured. A brief description of the characteristics of each site follows.

Rubidoux, CA, (RUB) is in the northwest corner of Riverside County. It is subject to pollutants transported from the Los Angeles basin, and also has local agricultural pollutant sources. It has historically had high levels of PM_{10} and ammonium nitrate.[10,11]

Long Beach, CA, (LB) is adjacent to and south of Los Angeles. It is a coastal area, and subject to both the Los Angeles urban plume and marine air influences.

Atascadero (ATA) is in San Luis Obispo County, CA, well to the north of the Los Angeles basin. It is a small town in a rural area with few dominant local sources of PM_{10}, except for wood smoke during the winter.

Pedregal (PED) is a residential area within Mexico City, southwest of downtown, that has few local sources, except for vehicular emissions. Prevailing daytime winds transport pollutants from the city center and industrial areas north of the city to this site.

Merced (MER) is in downtown Mexico City, approximately 1 km northwest of the airport. The area is close to a major highway in a commercial and residential zone.

Tlalnepantla (TLA) is in northwest Mexico City, about 12 km from downtown. The site is located in an industrial zone.

Washington, D.C., (DC) has relatively few industrial sources for a large urban area; vehicular sources and transported regional pollutants make up much of the $PM_{2.5}$. Measurements were made in the District, 2 miles north of the U.S. Capitol.

Philadelphia, PA, (PHL) is similar to D.C., with vehicular and regional sources being the dominant contributors to $PM_{2.5}$ concentrations. Measurements were made in the northeast area of the city, in a suburban residential area.

Boston, MA, (BOS) is a coastal site, with measurements made about 1 mile west of Boston Harbor in South Boston. Sources of particulate mass include industrial and vehicular sources, transported pollutants, and possibly marine aerosols.

Uniontown (UNT) is a town in rural southwest Pennsylvania near several large industrial sources along the Ohio and Monongahela rivers. Summer aerosol is dominated by sulfate, with some organic material from local open burning.

Table 2. Sampling locations.

Site	Season	Size Fraction
Long Beach, CA	All seasons	PM_{10}
Rubidoux, CA	All seasons	PM_{10}
Atascadero, CA (semi-rural)	All seasons	PM_{10}
Mexico City (Pedregal)	Winter	PM_{10}
Mexico City (Merced)	January-June	PM_{10}
Mexico City (Tlalnepantla)	January-June	PM_{10}
Philadelphia PA	Summer	PM_{10} and $PM_{2.5}$
Washington DC	Summer	PM_{10} and $PM_{2.5}$
Boston MA	Winter	PM_{10} and $PM_{2.5}$
Uniontown PA (semi-rural)	Summer	$PM_{2.1}$

Data Analysis

Means of TEOM data for matching 24-hour periods were regressed on the time-integrated (manual) method data for each site. For sites with data from multiple seasons, data were broken into warm and cold weather periods. Means were calculated for each method at each site.

RESULTS

Table 3 summarizes the results from the comparisons between the methods for both PM_{10} and $PM_{2.5}$ at the sampling sites. It lists the sample size, regression parameters, the mean of each method, and the ratios of the means, as well as the face velocities at the TEOM and manual method filters. For the purpose of comparing $PM_{2.5}$ and PM_{10} data, the ratio of PM_{10} means are also shown without the $b_0 = 3$ µg m^{-3} and $b_1 = 1.03$ calibration factors in a parenthesis.

The PM_{10} methods agree within 15% for samples collected during warm weather seasons at all sites except those in Rubidoux and Mexico City. For the Washington and Philadelphia sites, the correlation coefficient (r^2) was 0.94 and 0.97, respectively, with ratios of means (TEOM to manual method) of 1.06 and 0.96, respectively. For the cold seasons, the agreement between methods was not as good at any site. Boston and Mexico City had r^2 values of 0.84 and 0.61, respectively, with ratios of means 0.92 and 0.82, respectively. The largest differences between the overall PM_{10} means were observed at Rubidoux, CA, and the three Mexico City sites. These differences were independent of the sampling period. The correlation r^2 for Rubidoux was 0.59 for summer and 0.86 for winter, whereas the ratio of means was 0.79 for summer and 0.73 for winter. TEOM and manual method were relatively poorly correlated in any Mexico City site (r^2 ranged from 0.27 to 0.61), whereas the ratio of the TEOM to manual PM_{10} means ranged from 0.62 to 0.82.

For Atascadero, CA, (a semi-rural site) wood smoke is expected to be the primary cause of difference between the methods during winter, causing the ratio of the means to be 0.84 ($r^2 = 0.80$). The agreement between TEOM and time-integrated methods in this site during the summer is excellent (means ratio is 1.04). Figure 1 shows a plot of the difference between the methods at Atascadero, CA, from December 1993 through November 1994. The cold weather months show a consistent difference between the methods (TEOM less than SSI HiVol), with a smaller and more random difference during warmer months.

For $PM_{2.5}$, a reasonably good agreement was observed between TEOM and manual methods during summer in Philadelphia, Washington, and Uniontown, with the TEOM mean concentrations somewhat lower than the manual methods. The r^2 values range from 0.87 to 0.97, with ratios of means from 0.84 to 0.89. Larger differences between TEOM and time-integrated sampling methods were observed in Boston, particularly during winter sampling. The ratio of the TEOM vs. manual methods means was 0.64 and 0.78 for winter and summer, respectively. The absolute values for concentrations were also the lowest at this site, with means in the range of 8-12 µg m^{-3} for both TEOM and manual methods. For both sampling periods, the concentrations measured by TEOM and manual methods were highly correlated (r^2 was 0.99 and 0.87 for summer and winter, respectively).

DISCUSSION

There is much variability in the agreement between the TEOM and time-integrated sampling methods for PM_{10}. Except for Rubidoux and Mexico City, the data collected

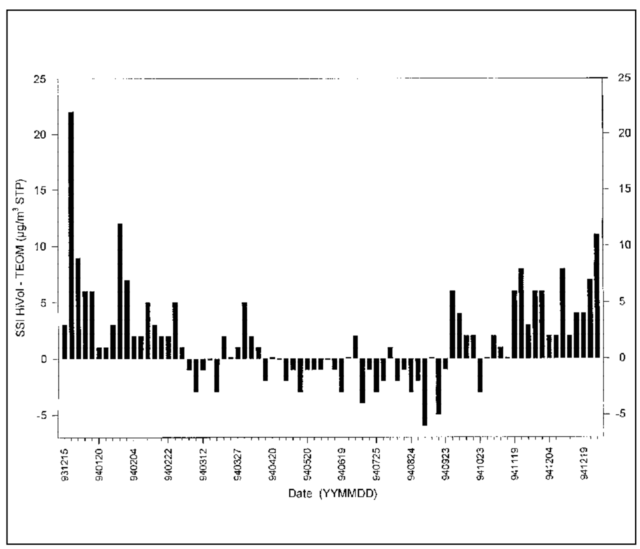

Figure 1. Difference in the 24-hour average PM_{10} concentrations between the Sierra-Anderson High-Volume sampler and the TEOM at Atascadero, CA, as a function of date.

Table 3. Comparisons of TEOM vs. manual PM methods.

		PM	**Manual**	**Sample**				**- - - - Means - - - - -**		**Face Velocities****
Site	Season	Size	Method	Size	r^2	b_0	b_1	TEOM/Manual	Ratio	TEOM/Manual
RUB	summer	10 mm	SSI	31	0.59	13*	0.62	61 / 77	0.79	38 / 41
RUB	winter	10	SSI	30	0.86	12	0.51	39 / 54	0.73	38 / 41
LB	summer	10	SSI	30	0.90	-1*	0.89	32 / 37	0.85	38 / 41
LB	winter	10	SSI	28	0.81	8	0.58	31 / 41	0.76	38 / 41
ATA	summer	10	SSI	30	0.85	2*	0.96	22 / 21	1.04	38 / 41
ATA	winter	10	SSI	48	0.80	4	0.68	18 / 22	0.84	38 / 41
TLA	Jan-Jun	10	SSI	31	0.27	15	0.46	71 / 115	0.62	38 / 41
MER	Jan-Jun	10	SSI	31	0.35	-2	0.81	73 / 92	0.79	38 / 41
PED	winter	10	4 lpm HI	49	0.61	8	0.69	53 / 64	0.82	38 / 8
BOS	winter	10	HI	31	0.84	3	0.74	17 / 18	0.92	38 / 20
BOS	summer	10	HI	45	0.95	0*	1.03	21 / 20	1.05	38 / 20
DC	summer	10	HI	27	0.97	3	0.93	29 / 28	1.06	38 / 20
PHL	summer	10	HI	74	0.94	0*	0.97	25 / 26	0.96	38 / 20
BOS	winter	2.5	HI	30	.87	0*	0.64	8 / 12	0.64	45 / 20
BOS	summer	2.5	HI	45	.99	-2	0.94	10 / 12	0.78	45 / 20
DC	summer	2.5	HI	24	.87	-4	0.96	17 / 21	0.84	23 / 20
PHL	summer	2.5	HI	43	.97	-2	0.97	17 / 19	0.89	50 / 20
UNT	summer	2.1	HEADS	61	.96	+1*	0.87	28 / 33	0.84	50 / 12

* Not significant at p = .05 ** Sampler filter face velocity in cm/sec

1. $PM_{2.5}$ TEOM data do not include gain and offset factors as is required for PM_{10}. The "raw" ratio for PM_{10} is also without the +3% and +3 µg/m^3 factors for direct comparison with fine mass TEOM data.
2. The manual PM method (SSI or HI) is considered the independent variable for these regressions.
3. "Summer" data are from May through October for all sites except DC and PHL, which were June through August.
4. Harvard Impactor (HI) and HEADS sampler flows are 10 lpm, except as noted. TEOM filter flows range from 1.8 to 4.0 lpm, depending on the inlet flow and whether an inlet bypass flow was used or not.

during the summer by both sampling methods agree reasonably well for PM_{10}. In the winter, the agreement is not as good at any site. The largest differences between PM_{10} methods were observed at the Rubidoux site and the three Mexico City sites. These were also the sites where correlation between the methods was minimal. Because of the location of the Mexico City sites, PM composition is expected to be dominated by vehicular contributions and industrial emissions. Thus, particles at these sites should contain a large fraction of semi-volatile compounds. This could explain the differences between the TEOM and manual methods.

A scatter plot of the Rubidoux data is shown in Figure 2. Note that on the two highest concentrations, the SSI/TEOM ratio exceeds a factor of 2. Figure 3 shows this difference as a function of date. The largest differences occur during the cooler months of December through early April, but the time of year with the most consistent signed difference between the methods is late April through mid-November. Rubidoux has relatively high concentrations of ammonium nitrate.[10,11] Most of the difference between the PM_{10} methods at the Rubidoux site can be accounted for on a sample-by-sample basis by the measured amount of ammonium nitrate on the SSI HiVol filter. To demonstrate this, the ammonium nitrate concentration measured on the SSI filter was added to the PM_{10} concentration measured with the TEOM, and the sum was plotted against the PM_{10} concentration of the SSI sampler. The resulting relationship is graphically shown in Figure 4. The slope of the linear regression is 1.01 (±0.04), and the intercept is not significantly different from zero, thereby suggesting that the entire difference between TEOM and SSI PM_{10} concentrations can be attributed to the ambient particulate ammonium nitrate that was lost to volatilization upon collection on the TEOM filter. The ratio of the TEOM to manual method means was also substantially smaller than

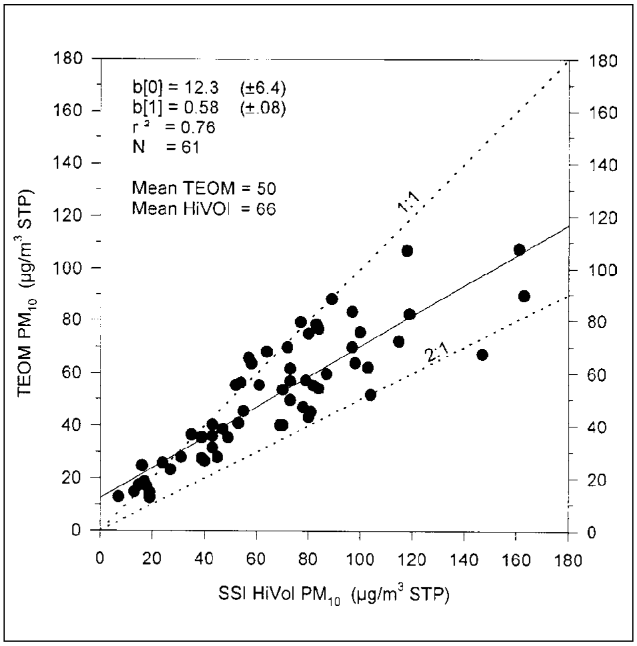

Figure 2. Plot of the 24-hour average PM_{10} concentrations determined with the TEOM vs. the 24-hour average PM_{10} concentrations determined with the Sierra-Anderson High-Volume sampler at Rubidoux, CA.

1.0 at Long Beach in the Los Angeles basin (0.85 and 0.76 for summer and winter, respectively), where ammonium nitrate levels are also known to be high.[11] These results do not imply that the SSI HiVol method does not lose any ammonium nitrate, but they do suggest that the TEOM does not retain any of this semi-volatile PM component.

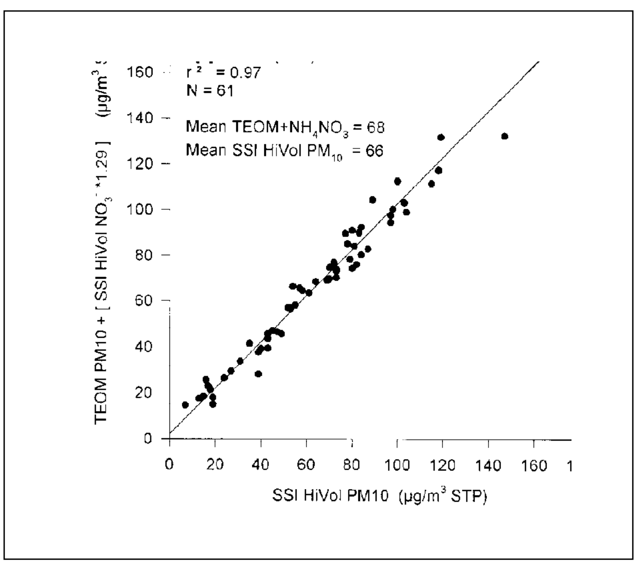

Figure 4. Plot of the sum of the 24-hour TEOM PM_{10} and ammonium nitrate concentrations vs. the 24-hour average PM_{10} concentrations determined with the Sierra-Anderson High-Volume sampler at Rubidoux, CA.

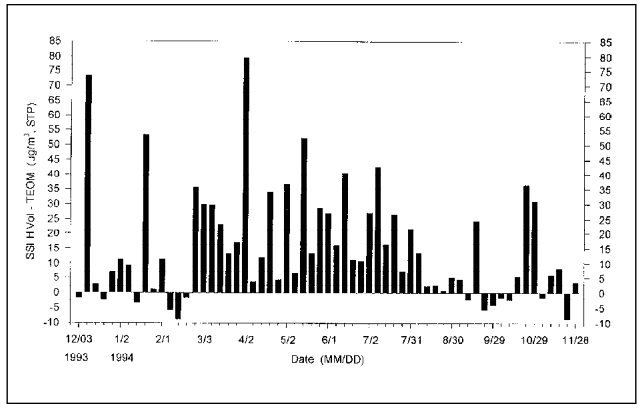

Figure 3. Difference in the 24-hour average PM_{10} concentrations between the Sierra-Anderson High-Volume sampler and the TEOM at Rubidoux, CA, as a function of date.

All of the TEOM $PM_{2.5}$ data are lower than the corresponding manual method data, regardless of season. This is consistent with the loss of semi-volatile mass and water associated with the aerosol, which are both primarily in the fine size fraction.[13] The lowest value of the ratio of the means between TEOM and manual methods was observed during the winter in Boston. The fine aerosol mass during the winter in the urban northeast United States is often dominated by local vehicular sources, and therefore is expected to have a strong component of semi-volatile organic material. Day to day variations in the composition of the fine aerosol mass would produce differences in the measured TEOM mass concentration relative to the manual methods.

Because of the accumulation of fine mass on the TEOM filter over its approximately two week lifetime, it is possible that adsorption or desorption of material occurs on the filter when the air pollutant concentrations change. For instance, Figure 5 shows the TEOM PM_{10}, TEOM $PM_{2.5}$, and black aerosol carbon (as measured by light absorption on a filter) in 15-minute increments at the Boston site on February 19, 1995. Instrument malfunction was initially suspected to be the cause of the large irregularities shown here; however, this was ruled out because of the good temporal correlation of the TEOM fluctuation with the other particle parameters, such as observed weather conditions and internal instrument readings. The TEOM $PM_{2.5}$ filter loading signal is shown on a relative 0-100% scale; this signal represents the relative pressure drop across the TEOM filter (assuming flow is constant) and increases slightly with sampling duration because of particle loading on the filter. Thirty-minute average values for the total particulate loading of the TEOM filter (divided by a factor of 4 to get it on scale on this plot) are also shown.

A change in the air mass composition occurred between 6:00 a.m. and 7:00 a.m., as indicated by the

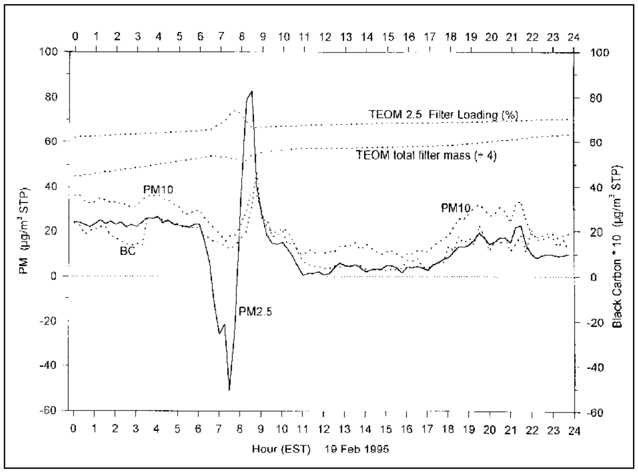

Figure 5. TEOM PM_{10} and $PM_{2.5}$ concentrations as well as black carbon (BC) concentration as a function of time at the Boston site on February 19, 1995. Also shown are the percentage loading of the TEOM filter and the total TEOM filter mass (divided by 4). BC concentrations are multiplied by a factor of 10 for scaling. All BC and TEOM concentrations represent averages of 15 minutes.

moderate decreases in TEOM PM_{10} and black carbon concentrations. The TEOM $PM_{2.5}$ concentration dropped dramatically, reaching a 15-minute mean of about -50 µg m^{-3} at 7:30 a.m. PM_{10} and black carbon levels briefly peaked just before 9:00 a.m., reaching levels somewhat higher than they had been 5 hours previously. The TEOM $PM_{2.5}$ concentration during this time increased rapidly, peaking at 80 µg m^{-3} just prior to the PM_{10} and black carbon peak. There was no indication of TEOM $PM_{2.5}$ malfunction during this time period, and the total PM loading of the TEOM filter indicates a change in the same direction indicated by the concentration data, ruling out flow problems. This suggests that the wide excursions in the TEOM $PM_{2.5}$ concentration data were not instrument malfunctions, but artifacts associated with changes in the ambient composition. Photographic data for 8:00 a.m. on this date show a clearly delineated fog bank within a few miles of the site, which is located less than one mile from Boston Harbor. Rapid changes in the composition of the air mass (urban air vs. marine air), accompanied by large changes in dew point as the fog bank shifts on and off the coast, are likely to have been the cause of these fluctuations in TEOM $PM_{2.5}$ data.

This type of response to changes in sample composition is not usually observed. However, it is likely that similar, but smaller, effects could occur frequently. The PM_{10} TEOM did not show any indication of instability during this period, as evidenced by the close tracking of PM_{10} and black carbon between 6 a.m. and 11 a.m. One possible explanation for the stability of PM_{10} measurements is that the coarse fraction particles (about 40-50% of the PM_{10}) somehow stabilize, either physically or chemically, the semi-volatile particulate matter to some extent once coarse and fine particles are collected simultaneously on the TEOM filter.

It is important to note that throughout this discussion any integrated sampling method for PM, including the FRM, is subject to some loss of semi-volatile particulate mass. The data presented here simply indicate that the TEOM method usually loses more of this mass component than the FRM or similar manual methods, and therefore gives a lower measurement of PM in areas where semi-volatile mass is a substantial component of the PM.

CONCLUSIONS

Monitored sites and seasons expected to have high levels of ammonium nitrate and/or organic particulate mass showed poor correlation between the PM_{10} and $PM_{2.5}$ methods, with the TEOM data being generally lower than the manual methods. Seasonality must be taken into account; the same site may show good agreement in summer and poor agreement in winter.

The relationship between traditional gravimetric SSI HiVol PM_{10} methods and the TEOM PM_{10} varies widely based on location of site, time of year, and range of particle concentrations. Concentrations of ammonium nitrate and condensed-phase semi-volatile organics are retained on the TEOM filter for only a brief time, causing mass data to be lower than SSI HiVol samples in areas where these compounds are a substantial fraction of the PM_{10}.

Since the difference in these PM_{10} methods is likely to be largest in heavily populated urban areas with high particle levels, state and local air monitoring agencies may need to address the differences for compliance purposes. Health effect and modeling research may also be needed to take these differences into account.

When used to sample $PM_{2.5}$, large fluctuations can occasionally occur in the TEOM apparent mass concentration over several hours. This noise is likely caused by changes in the equilibrium of particles on the TEOM filter when ambient pollutants or moisture is changing rapidly. PM_{10} TEOM data do not seem to be as significantly affected under these conditions.

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